

## Lecture 10: Power Laws and LOB Shape

Lecturer: Andrew Papanicolaou

Scribes: Daniel Chertock

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### 10.1 Microstructure Foundations: Deriving $f(\nu)$ from LOB Depth

In Lecture 1, we posited an exogenous temporary price impact function  $f(\nu)$  and defined the continuous cost rate as  $L(\nu) = \nu f(\nu)$ . We now establish the microstructural origin of  $f(\nu)$  by modeling the geometry of the Limit Order Book (LOB).

Let  $S_t$  denote the unaffected mid-price. Define  $x = P - S_t \geq 0$  as the price offset above the mid-price. The local liquidity profile of the order book is characterized by the limit order density function  $\rho(x) \geq 0$ , where  $\rho(x) dx$  represents the number of shares available for execution in the price interval  $[S_t + x, S_t + x + dx]$ .

**1. Clearing Order Book Depth:** To execute sell orders at a continuous speed  $\nu_t$ , or buy orders absorbing liquidity, the trade consumes limit orders up to a marginal price displacement offset  $\delta = f(\nu_t)$ . Over a characteristic order book replenishment time scale  $\tau > 0$ , the cumulative volume  $V(\delta)$  absorbed up to price offset  $\delta$  satisfies

$$V(\delta) = \int_0^\delta \rho(x) dx . \quad (10.1)$$

Balancing the continuous execution volume rate  $\tau \nu_t$  against the absorbed order book volume  $V(f(\nu_t))$  yields the fundamental microstructural relation

$$\int_0^{f(\nu_t)} \rho(x) dx = \tau \nu_t . \quad (10.2)$$

Inverting equation (10.2) expresses the marginal temporary price impact  $f(\nu_t)$  directly as a function of the order book density profile  $\rho(x)$ .

### 10.2 Power-Law LOB Profiles & Empirical Square-Root Impact

We parameterize the local limit order density near the mid-price using a general power-law profile with shape exponent  $\gamma > -1$  and density coefficient  $c > 0$ :

$$\rho(x) = cx^\gamma . \quad (10.3)$$

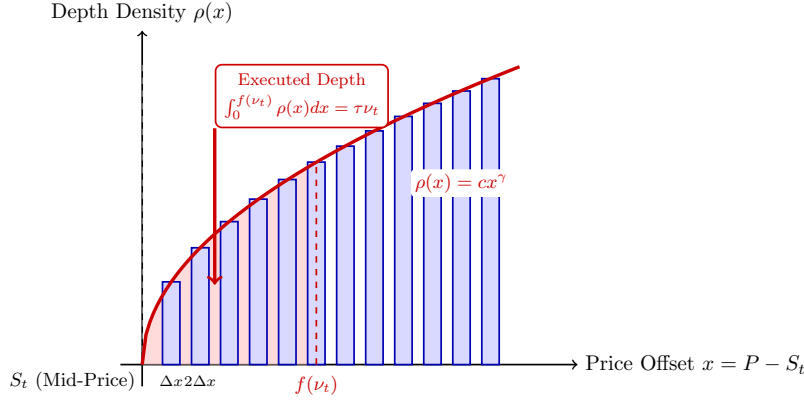


Figure 10.1: The power law for liquidity in an order book.

**1. Derivation of Power-Law Temporary Impact  $f(\nu)$ :** Evaluating the integrated volume in equation (10.2) under power-law density yields

$$\int_0^{f(\nu_t)} cx^\gamma dx = \frac{c}{\gamma+1} (f(\nu_t))^{\gamma+1} = \tau\nu_t. \quad (10.4)$$

Solving explicitly for  $f(\nu_t)$  gives

$$f(\nu_t) = \left( \frac{\tau(\gamma+1)}{c} \right)^{\frac{1}{\gamma+1}} \nu_t^{\frac{1}{\gamma+1}}. \quad (10.5)$$

Defining the impact exponent  $p := \frac{1}{\gamma+1} > 0$  and the liquidity parameter  $k := \left( \frac{\tau(\gamma+1)}{c} \right)^p > 0$ , the temporary price impact function reduces to the canonical power law

$$f(\nu_t) = k\nu_t^p. \quad (10.6)$$

Consequently, the continuous execution cost rate  $L(\nu_t) = \nu_t f(\nu_t)$  takes the non-quadratic form

$$L(\nu_t) = k\nu_t^{1+p}. \quad (10.7)$$

## 2. Special Cases of Order Book Geometry:

- **Flat LOB ( $\gamma = 0$ ):** Order density is uniform ( $\rho(x) = c$ ). This yields linear impact  $p = 1 \implies f(\nu) = k\nu$  and quadratic cost  $L(\nu) = k\nu^2$ , recovering the classic Almgren-Chriss specification.
- **Linearly Increasing LOB Depth ( $\gamma = 1$ ):** Order density increases linearly away from the mid-price ( $\rho(x) = cx$ ). This yields  $p = \frac{1}{2}$ , leading to the empirical **Square-Root Law** of price impact:

$$f(\nu) = k\sqrt{\nu} \implies L(\nu) = k\nu^{1.5}. \quad (10.8)$$

- **Sparse / Concentrated LOB ( $\gamma < 0$ ):** Liquidity is heavily clustered near the mid-price and thins out rapidly, producing super-linear impact ( $p > 1$ ).

## 10.3 Stochastic Control with Non-Quadratic Impact

Following the framework of Rogers and Singh (2010), we consider a risk-neutral agent liquidating  $Q_0 = \mathfrak{N}$  shares over  $[0, T]$  facing general power-law cost rate  $L(\nu) = k\nu^{1+p}$  ( $p > 0$ ). The value function  $H(t, S, q)$  represents maximum expected net execution proceeds:

$$H(t, S, q) = \sup_{\nu \in \mathcal{A}} \mathbb{E}_{t, S, q} \left[ \int_t^T (S_s \nu_s - k \nu_s^{1+p}) ds \right], \quad (10.9)$$

subject to  $dQ_s = -\nu_s ds$ , initial inventory  $Q_t = q$ , and terminal constraint  $Q_T = 0$ .

**Step 1: Value Function Decomposition and HJB PDE** Decomposing paper wealth into mark-to-market value  $qS$  plus execution value relative to paper mark,  $H(t, S, q) = qS + h(t, q)$ , the HJB equation for  $h(t, q)$  simplifies to

$$\frac{\partial h}{\partial t} + \max_{\nu} \left\{ -k\nu^{1+p} - \nu \frac{\partial h}{\partial q} \right\} = 0. \quad (10.10)$$

**Step 2: First-Order Condition for Non-Linear Speed** Differentiating the expression inside the maximum operator with respect to  $\nu$  gives

$$-(1+p)k\nu^p - \frac{\partial h}{\partial q} = 0 \implies \nu^*(t, q) = \left( -\frac{1}{(1+p)k} \frac{\partial h}{\partial q} \right)^{\frac{1}{p}}. \quad (10.11)$$

Substituting  $\nu^*$  back into equation (10.10) yields the non-linear PDE

$$\frac{\partial h}{\partial t} + pk \left( -\frac{1}{(1+p)k} \frac{\partial h}{\partial q} \right)^{\frac{1+p}{p}} = 0. \quad (10.12)$$

**Step 3: Power-Law Ansatz and ODE Reduction** Because cost rate  $L(\nu)$  is homogeneous of degree  $1+p$ , we propose the generalized non-quadratic ansatz

$$h(t, q) = -A(t)q^{1+p}. \quad (10.13)$$

Evaluating partial derivatives  $\frac{\partial h}{\partial t} = -\dot{A}(t)q^{1+p}$  and  $\frac{\partial h}{\partial q} = -(1+p)A(t)q^p$ , substitution yields

$$-\dot{A}(t)q^{1+p} + pk \left( \frac{(1+p)A(t)q^p}{(1+p)k} \right)^{\frac{1+p}{p}} = 0. \quad (10.14)$$

Factoring out  $q^{1+p}$  reduces the PDE to a separable ordinary differential equation for  $A(t)$ :

$$-\dot{A}(t) + pk^{-\frac{1}{p}} (A(t))^{\frac{1+p}{p}} = 0. \quad (10.15)$$

**Step 4: Solving for  $A(t)$  and Optimal Feedback Control** Separating variables and integrating yields

$$\int (A(t))^{-\frac{1+p}{p}} dA = pk^{-\frac{1}{p}} \int dt \implies -p(A(t))^{-\frac{1}{p}} = pk^{-\frac{1}{p}}(t - C). \quad (10.16)$$

Enforcing  $Q_T = 0$  requires execution cost to explode as  $t \rightarrow T$ , fixing  $C = T$ . Solving for  $A(t)$  gives

$$A(t) = \frac{k}{(T-t)^p}. \quad (10.17)$$

Substituting  $A(t)$  back into the feedback control yields the optimal trading speed

$$\nu^*(t, q) = \left( \frac{(1+p)A(t)q^p}{(1+p)k} \right)^{\frac{1}{p}} = \left( \frac{A(t)}{k} \right)^{\frac{1}{p}} q = \frac{q}{T-t} . \quad (10.18)$$

Integrating  $dQ_s^* = -\frac{Q_s^*}{T-s} ds$  from 0 to  $t$  gives the optimal trajectory and speed:

$$Q_t^* = \mathfrak{N} \left( 1 - \frac{t}{T} \right) \quad \text{and} \quad \nu_t^* = \frac{\mathfrak{N}}{T} . \quad (10.19)$$

## 10.4 Economic Interpretations & Options Hedging Link

**1. Universality of TWAP under Risk Neutrality:** Equation (10.19) demonstrates a powerful structural result: for *any* convex power-law impact cost  $L(\nu) = k\nu^{1+p}$  ( $p > 0$ ), a risk-neutral agent always executes at a constant speed  $\nu_t^* = \frac{\mathfrak{N}}{T}$ . By Jensen's inequality, constant trading rate minimizes the integrated cost  $\int_0^T L(\nu_t) dt$  for any strictly convex function  $L(\cdot)$ .

**2. Scaling of Execution Costs with Horizon  $T$ :** The total expected execution cost  $EC^* = -h(0, \mathfrak{N}) = A(0)\mathfrak{N}^{1+p}$  evaluates to

$$EC^* = \frac{k\mathfrak{N}^{1+p}}{T^p} . \quad (10.20)$$

This reveals how order book structure fundamentally alters horizon sensitivity:

- Under linear impact ( $p = 1$ ), doubling execution horizon  $T$  reduces total cost by 50% ( $EC^* \propto T^{-1}$ ).
- Under square-root impact ( $p = 0.5$ ), doubling  $T$  reduces total cost by only 29.3% ( $EC^* \propto T^{-0.5}$ ).

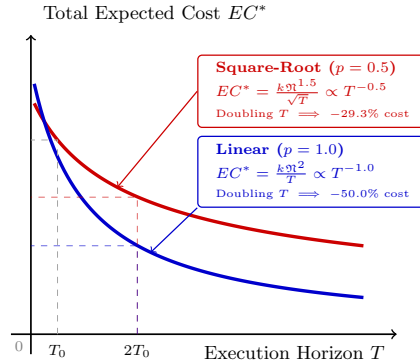


Figure 10.2: The cost decay of linear impact vs. square-root impact.

**3. Implication for Delta-Hedging Options (Rogers & Singh, 2010):** When continuous delta-hedging an option with Gamma  $\Gamma_t$ , rebalancing inventory requires trading volume rate  $\nu_t \propto |\Gamma_t \sigma S_t|$ . Under non-quadratic illiquidity  $L(\nu) = k\nu^{1+p}$ , the total expected cost of hedging scales non-linearly with option Gamma, forcing option market makers to adjust their implied volatility quotes by an illiquidity premium proportional to  $\Gamma_t^p$ .

## 10.5 Remarks

1. This lecture draws from [\[1\]](#).

## References

- [1] L. C. G. Rogers and S. Singh. The cost of illiquidity and its effects on hedging. *Mathematical Finance*, 20(4):597–615, 2010.